# Inferring Information: Part 1 (Structured Data)

*Making sense of what we don't understand*
*is somewhat equal to*
*trying to understand what seemingly doesn't make sense*

https://github.com/jendralhxr/introdatascience

# Better reports, what to consider: set out **objectives**

## Objective

1. To obtain monthly air pollution data from the official data.gov.my portal.

2. To visualize air pollution trends using Python.

3. To identify changes air pollution in Malaysia and determine the highest and lowest recorded value.

# Better reports, what to consider: **code** highlight

1.1 Import libraries

```
import matplotlib.pyplot as plt
import pandas as pd
import requests
```

1.2 Fetch data directly from data.gov.my API

```
API_URL =
"https://api.data.gov.my/data-catalogue?id=ridership_headline&date_star
t=2025-01-01@date&date_end=2025-12-31@date"

response = requests.get(API_URL)
data = response.json()
df = pd.DataFrame(data)
```

1.3 Data Processing (filter data for 2025)

```
df["date"] = pd.to_datetime(df["date"])
df["year"] = df["date"].dt.year
df_2025 = df[df["year"] == 2025]
df_2025.head()
```

```
import pandas as pd import matplotlib.pyplot as plt import requests # 1. Fetch Daily Public Transport Headline Ridership Dataset from data.gov.my URL_PARQUET = "https://storage.data.gov.my/transportation/ridership_headline.parquet" df = pd.read_parquet(URL_PARQUET) # 2. Data Preprocessing & Aggregation df['date'] = pd.to_datetime(df['date']) df['year'] = df['date'].dt.year # Filter for full annual cycles and sum trip totals across major rail networks df_yearly = df.groupby('year')[[ 'rail_lrt_kj', 'rail_mrt_kajang', 'rail_lrt_ampang', 'rail_monorail', 'rail_mrt_pjy' ]].sum().reset_index() # 3. Plotting Annual Rail Ridership Trends plt.figure(figsize=(12, 6), dpi=300) plt.plot(df_yearly['year'], df_yearly['rail_lrt_kj'] / 1e6, marker='o', color='#0284c7', linewidth=2, label='LRT Kelana Jaya') plt.plot(df_yearly['year'], df_yearly['rail_mrt_kajang'] / 1e6, marker='s', color='#16a34a', linewidth=2, label='MRT Kajang') plt.plot(df_yearly['year'], df_yearly['rail_lrt_ampang'] / 1e6, marker='^', color='#dc2626', linewidth=2, label='LRT Ampang') plt.plot(df_yearly['year'], df_yearly['rail_monorail'] / 1e6, marker='d', color='#d97706', linewidth=2, label='KL Monorail') plt.plot(df_yearly['year'], df_yearly['rail_mrt_pjy'] / 1e6, marker='x', color='#9333ea', linewidth=2, label='MRT Putrajaya') plt.title("Annual Rapid Rail Passenger Volume in Klang Valley (2019–2025)", fontsize=14, fontweight='bold') plt.xlabel("Year", fontsize=11) plt.ylabel("Annual Trips (Millions)", fontsize=11) plt.grid(True, linestyle='--', alpha=0.5) plt.legend(title="Rail Service Line", loc="upper left") plt.tight_layout() plt.show()
```

Use `monospace` font.
Color is nice, but stick to official color schemes.

# Better reports, what to consider: **data description**

*Table 1: Dataset information*

| Item | Description |
|---|---|
| Dataset name | Drug Addicts by State & Age Group |
| Dataset ID | drug_addicts_age |
| Data provider | National Anti-Drugs Agency (AADK) |
| Publisher | data.gov.my |
| Geographical coverage | Malaysia, including states and federal territories |
| Time coverage used | 2015–2025 |
| Frequency | Annual |
| File format | CSV and Parquet |
| Licence | Creative Commons Attribution 4.0 International (CC BY 4.0) |

*Table 2: Dataset variables*

| Variable | Description |
|---|---|
| date | Year of the data, represented as YYYY-01-01. |
| state | State where addicts were recorded; Malaysia represents the national total. |
| age_group | Age category: <13, 13–15, 16–19, 20–24, 25–29, 30–34, 35–39, 40+; total represents all age groups. |
| addicts | Number of drug addicts recorded for the specified year, state and age category. |

# Not too long of a **paragraph**

**5. Discussion**

The national trend indicates that recorded drug addiction figures changed over the selected period rather than following a steady increase or decrease. The decline observed between 2019 and 2021 was followed by a recovery in the recorded figures from 2022 to 2024, and a decrease in 2025. These movements describe the administrative records and do not, by themselves, establish the causes of the changes.

The age-group analysis provides a demographic perspective by showing how recorded cases are distributed across age categories. The state comparison adds a geographical perspective and may help identify where recorded cases are concentrated. The exact highest and lowest age groups and states should be interpreted from the generated 2025 tables and charts.

The findings may support planning for prevention, awareness, treatment and rehabilitation programmes. However, several limitations should be considered. First, the dataset may not include unreported cases. Second, the number of recorded addicts is a count and is not adjusted for the population size of each state; therefore, it should not be treated as a rate or a direct measure of risk. Third, administrative reporting and identification practices may differ across places and years. Further research could combine these data with state population estimates to calculate rates per 100,000 residents and with other social or public-health indicators.

**3.0 Discussion**

The graph shows that W.P. Kuala Lumpur recorded the highest mean gross monthly household income in 2024, at RM13,985. W.P. Putrajaya followed with RM13,846, while Selangor recorded RM13,296. These figures indicate that the mean household income was higher in these Federal Territories and state than in the other areas shown. Kelantan recorded the lowest mean income, at RM5,265, followed by Kedah at RM5,793 and Perlis at RM5,996. This shows that household income levels were not the same across Malaysia in 2024. The median income is lower than the mean income in every state and Federal Territory in the graph. This pattern is consistent with income distributions where a smaller number of households with very high incomes can raise the mean. The median is less affected by extremely high values and can therefore provide a useful indication of the middle household income. For example, W.P. Kuala Lumpur had a mean income of RM13,985 and a median income of RM10,802, giving a difference of RM3, 183. In Kelantan, the mean was RM5,265 and the median was RM4,083, a difference of RM1, 182. These differences show why it is useful to examine both measures instead of relying on the mean alone. Overall, the graph highlights differences in gross monthly household income across states and Federal Territories. However, the graph describes income levels only; it does not explain the causes of these differences. Other factors would need to be studied before making conclusions about why income varies between locations.

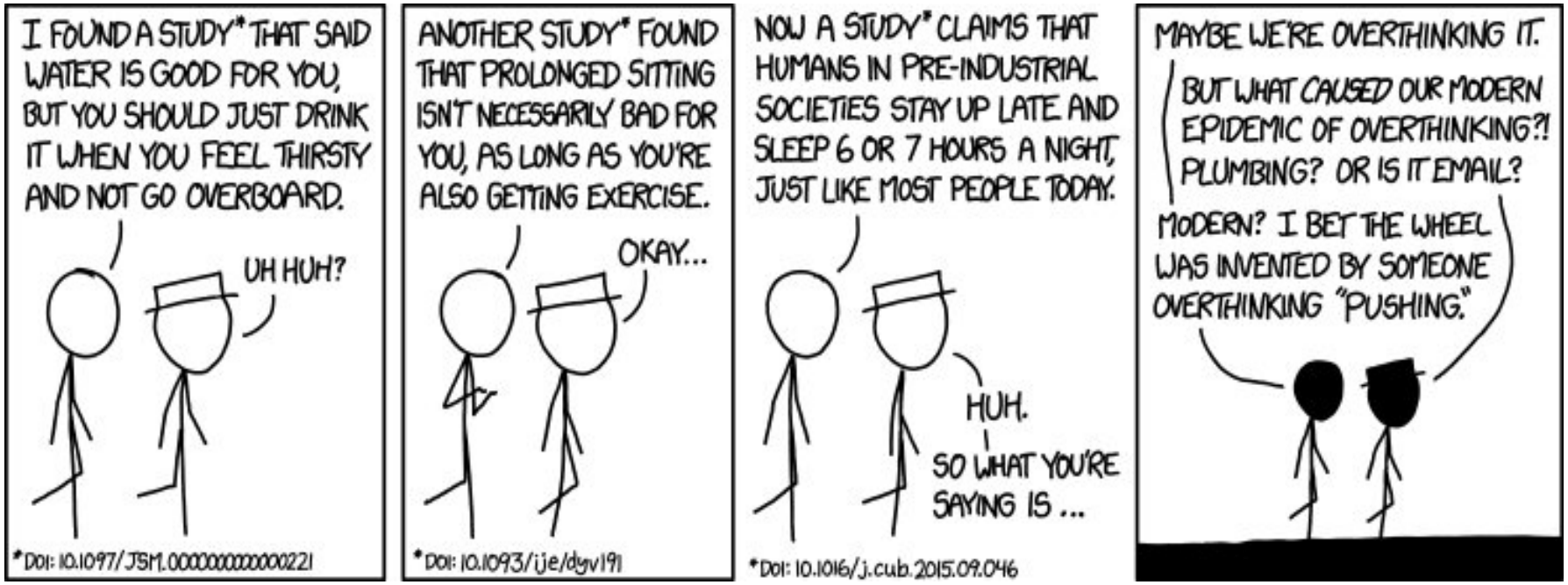
# Attendance List




https://forms.gle/q7yN33EmTYT5dpLB6

# Epistemology

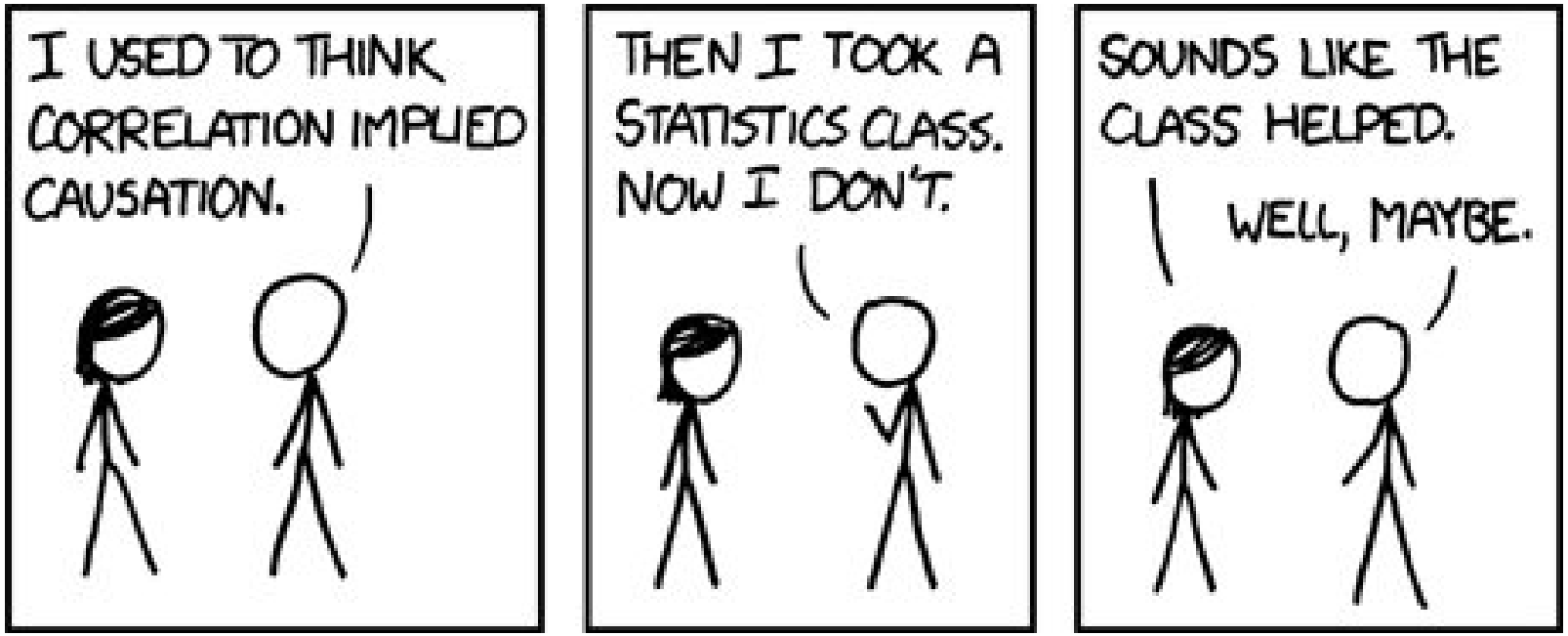
Statistical analysis and modern system engineering pushes human cognition exhibits a persistent bias toward narrative assembly, i.e. speculating causal linkages out of coincidental co-occurrences.



# Definition

Causation: *something* happens **because** *some other thing* happens in a particular way

Correlation: *something* is happening **along** *some other thing* happening




https://xkcd.com/552/

# What to observe then?

**Strength/scale**: how big is the effect? Due to how much dose/changes applied?
**Consistency**: is it *always* happening as expected?
**Specificity**: What is the actual cause?
**Temporality**: what happens first leading to what, *when*?
**Plausability**: does this even makes sense based on current knowledge/understanding?
**Coherence**: does this even makes sense based on series of logical evaluations?
**Analogy**: similar things happens elsewhere?




https://www.kdnuggets.com/2017/02/hill-data-scientist-xkcd-story.html

# A case from data.gov.my

- Population (how many people)

https://api.data.gov.my/data-catalogue?id=population_district

- Income (how much money to afford)

https://api.data.gov.my/data-catalogue?id=hh_income_district

- Hospital Beds available (healthcare capacity)

https://api.data.gov.my/data-catalogue?id=hospital_bed

```
df_pop.columns
Index(['state', 'district', 'date', 'sex', 'age', 'ethnicity', 'population',
       'year'], dtype='str')

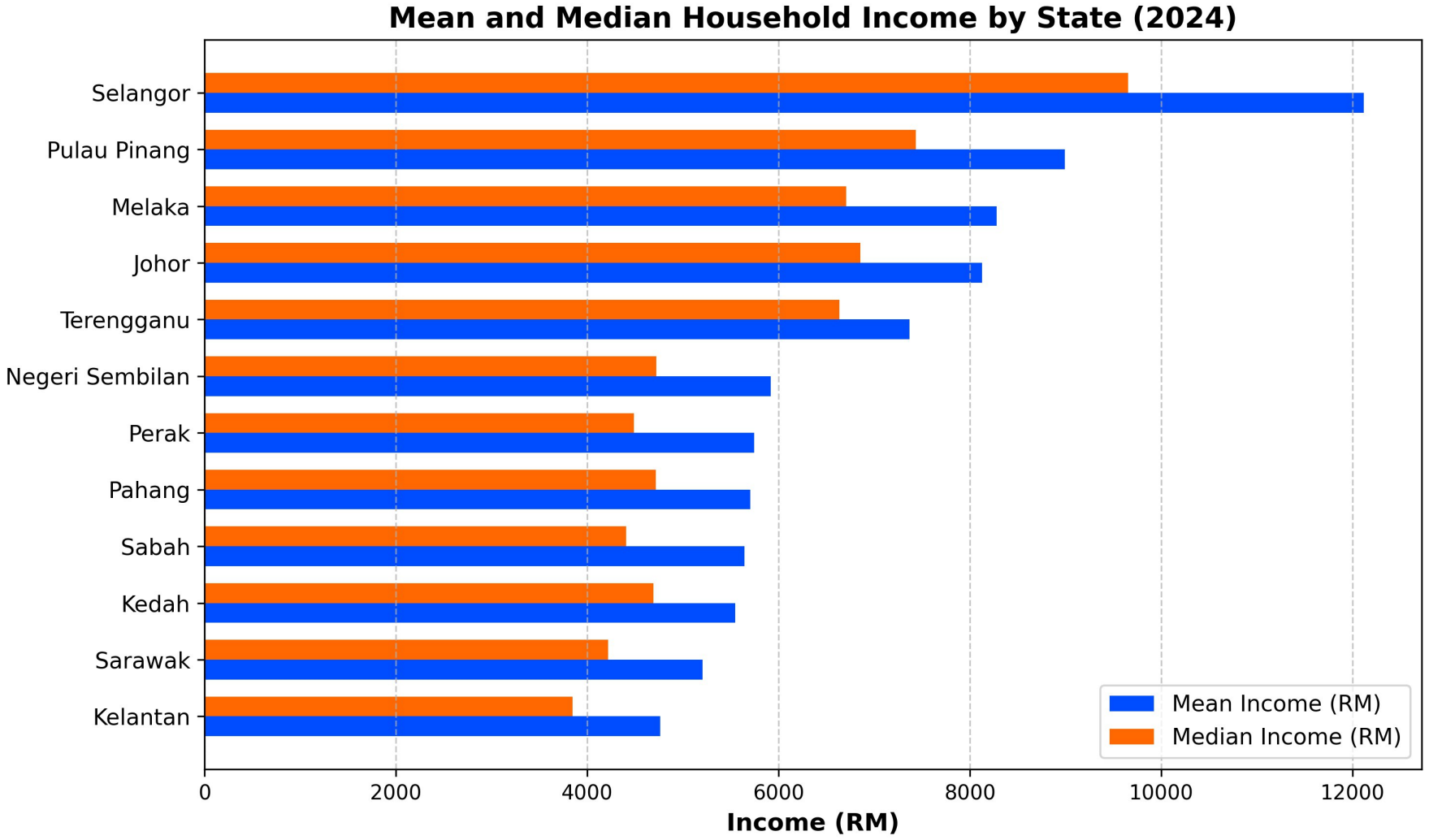
df_inc.columns
Index(['state', 'district', 'date', 'income_mean', 'income_median', 'year'], dtype='str')

df_beds.columns
Index(['date', 'state', 'district', 'type', 'beds', 'year'], dtype='str')
```

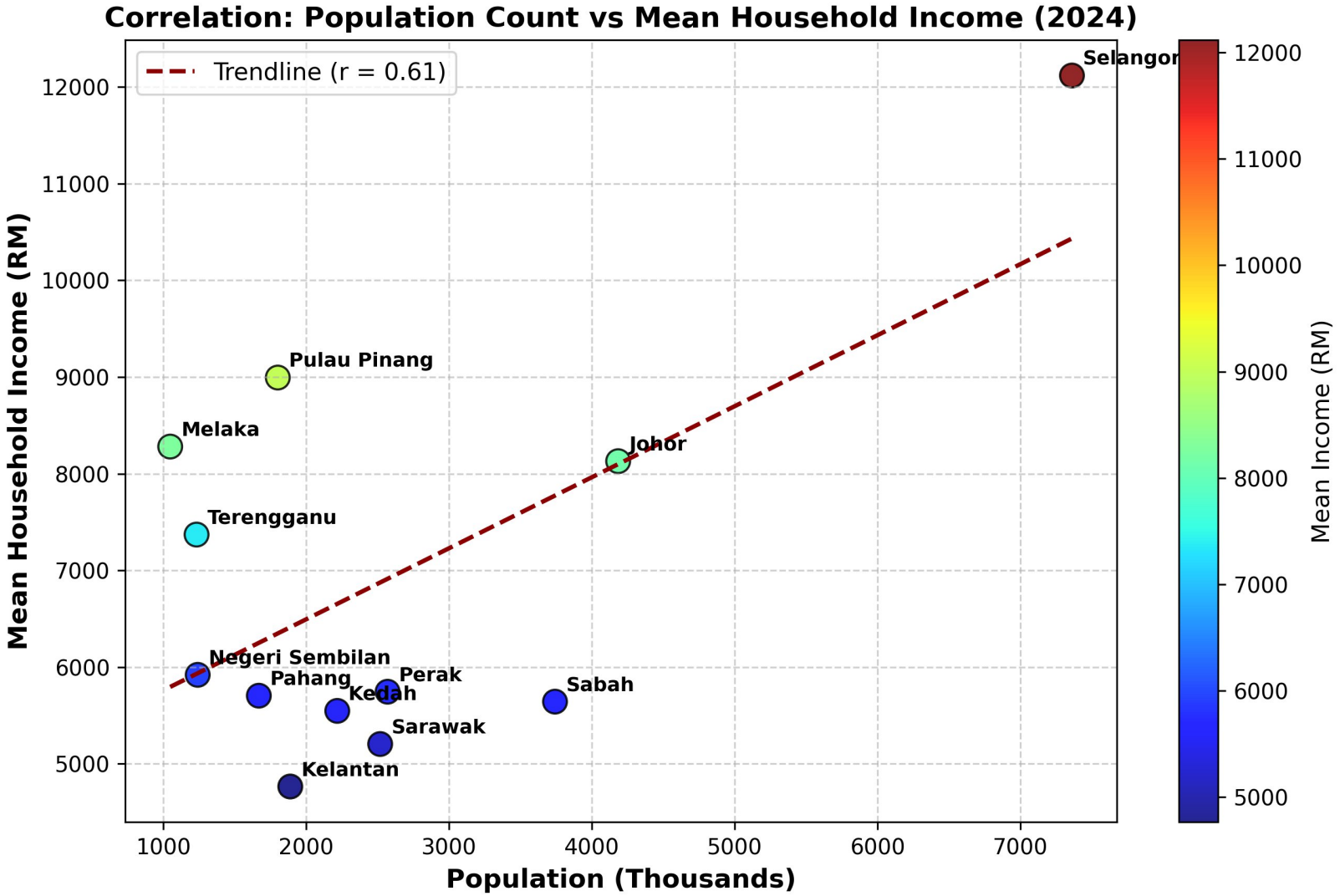
# Population

| | State | Population (Thousands) |
|---|---|---|
| 0 | Selangor | 7362.7 |
| 1 | Johor | 4184.3 |
| 2 | Sabah | 3742.1 |
| 3 | Perak | 2569.4 |
| 4 | Sarawak | 2517.4 |
| 5 | Kedah | 2217 |
| 6 | W.P. Kuala Lumpur | 2067.5 |
| 7 | Kelantan | 1887.8 |
| 8 | Pulau Pinang | 1800.4 |
| 9 | Pahang | 1667.6 |
| 10 | Negeri Sembilan | 1239.5 |
| 11 | Terengganu | 1231.5 |
| 12 | Melaka | 1046.8 |
| 13 | Perlis | 296.8 |
| 14 | W.P. Putrajaya | 120.3 |
| 15 | W.P. Labuan | 100.8 |

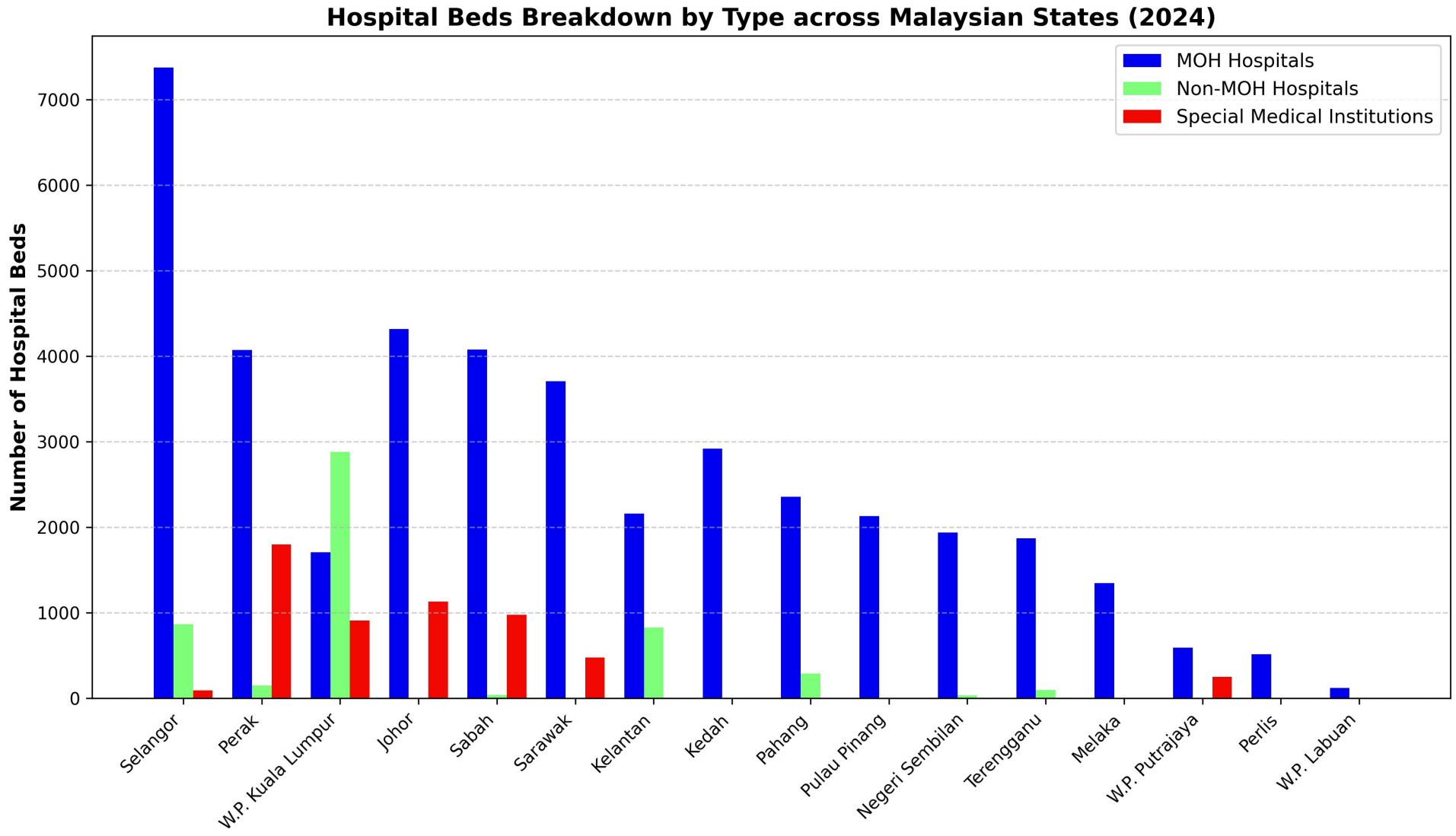
# Income


Mean and Median Household Income by State (2024)
Selangor
Pulau Pinang
Melaka
Johor
Terengganu
Negeri Sembilan
Perak
Pahang
Sabah
Kedah
Sarawak
Kelantan
0
2000
4000
6000
8000
10000
12000
Income (RM)
Mean Income (RM)
Median Income (RM)

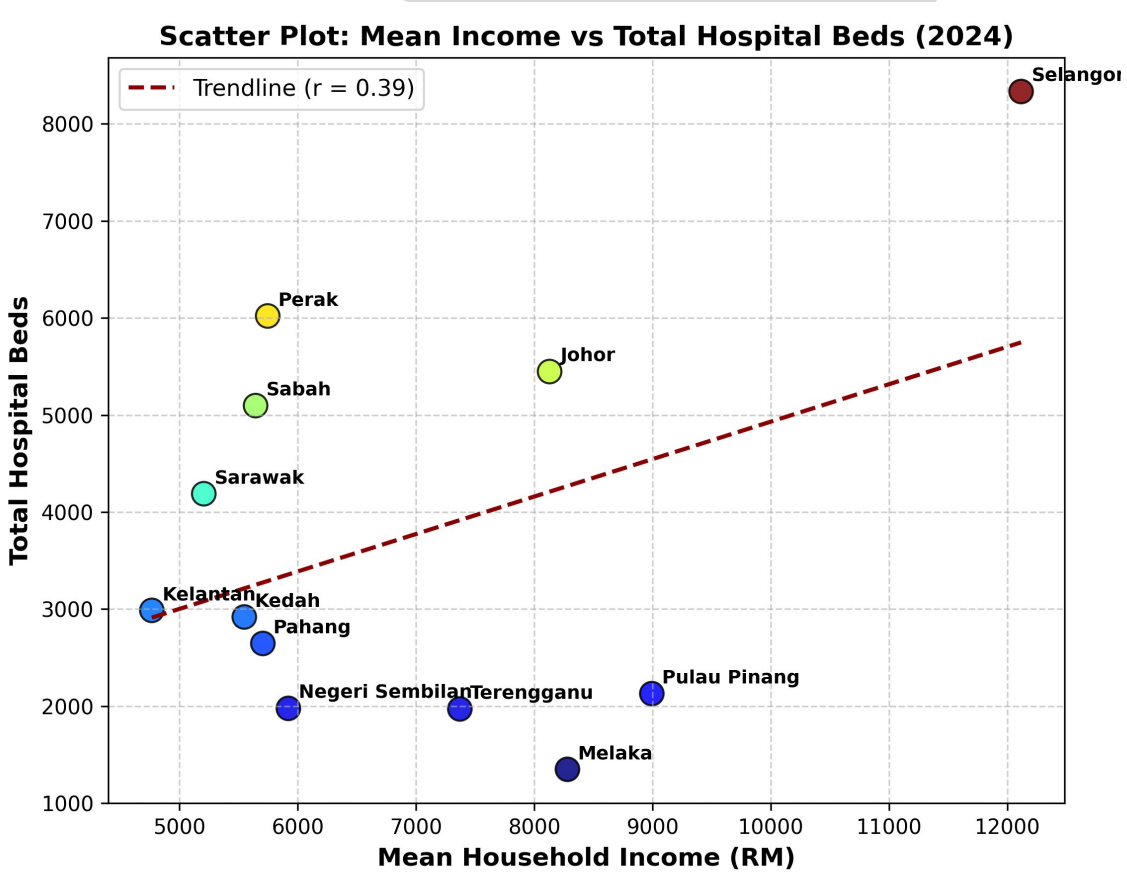
# Population-driven by income?

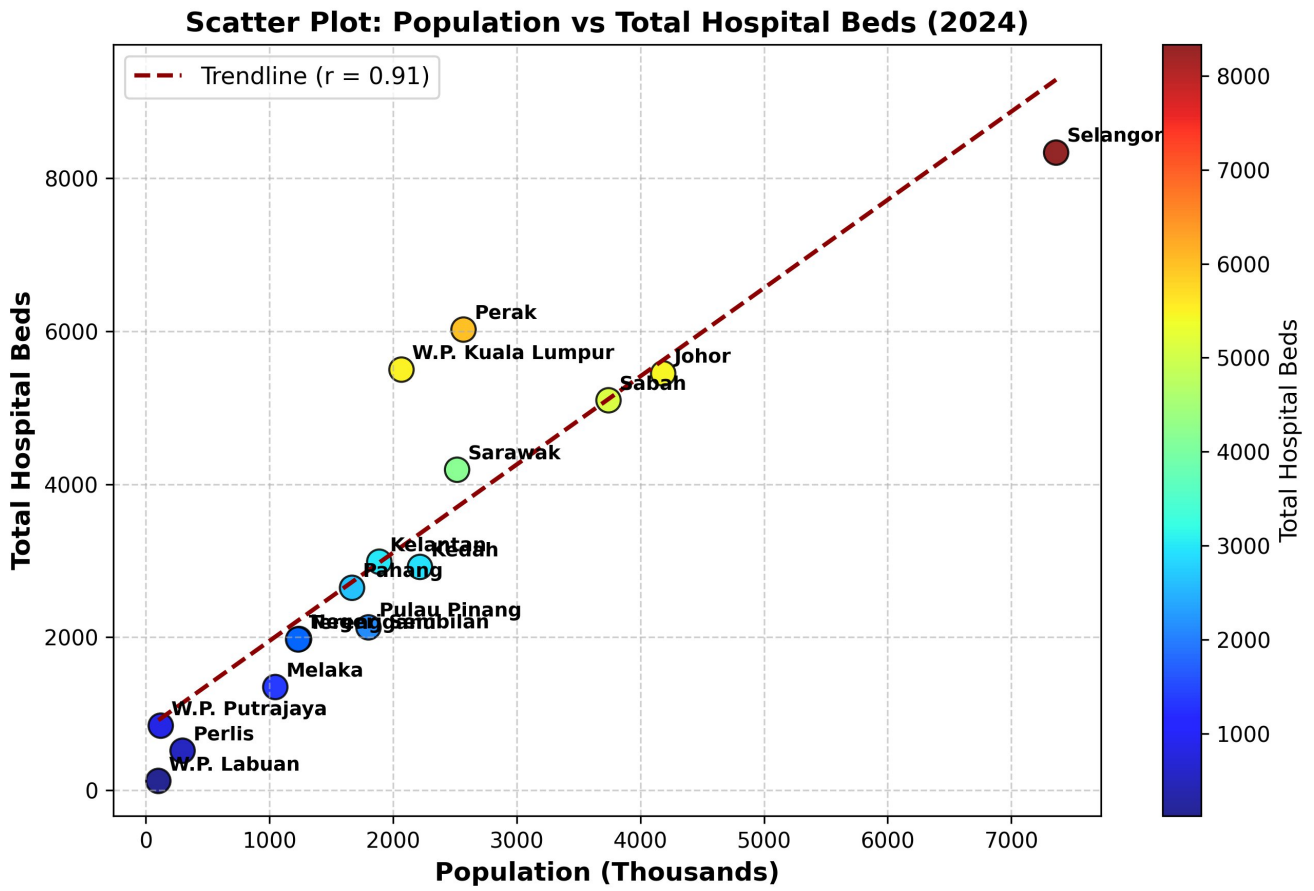


# Hospital beds



# Hospital beds availability



# Assignment or not?